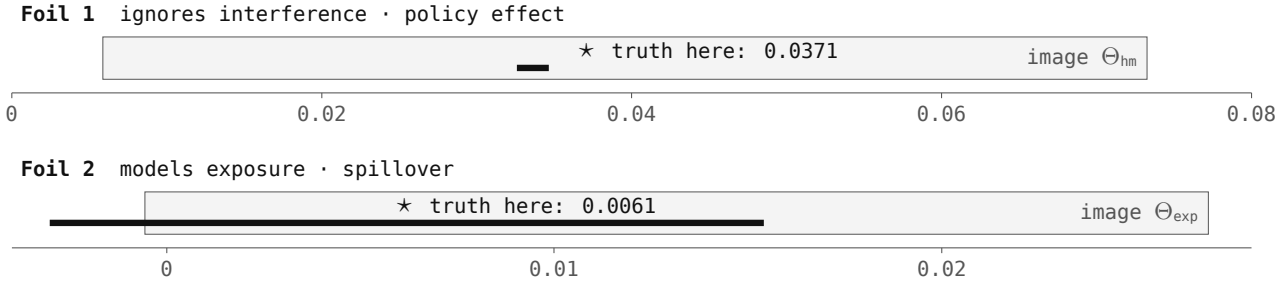
Not-ruled-out images of policy contrasts

A village agent-based model, calibrated by history matching, and two one-world analysts graded against what the targets cannot rule out

The paper in one picture



The diagram shows the min-I point, the best-fitting NROY point. The two rows use different scales. Bands: the image of the effect over 400 sampled NROY points. Black lines: the analyst's nominal 95% interval from one panel. Stars: the simulator's true value at this point. Foil 1 is tight, sits inside the band, and misses the truth. Foil 2 contains the truth, but it is too wide for a robust verdict.

The question

Many parameter points fit the same three targets, and they imply different campaign effects. What must a calibrated simulator report? How far can an interval computed at one point be trusted?

The answer

Foil 1 is F-OVERCONF at both diagnostic points; no robust Foil 2 verdict is claimed at either point. Most of Foil 1's miss is the effect on people who were not seeded. Report the image, not one point.

In numbers

Points evaluated in three history-matching waves / points kept (NROY)	3,000 / 991
Image of the policy effect over the sampled NROY points	[0.0059, 0.0732]
Policy effect at the planted point c*	0.0383
Foil 1 estimate below the truth, at min-I / median-I	9.3% / 7.2%
Foil 2 interval coverage when the exposure map holds exactly	93.4%

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Draft for review, October 2026
github.com/Bonorinoa/ecn593-applied-project
Deterministic · 52 tests · one command replicates it

A calibrated simulator gives a policy effect for each parameter point that fits the targets. Many points fit the targets equally well, yet they imply different effects. This note therefore reports the *image* of the policy effect over the sampled NROY points, not one number. The image follows the *reporting logic* of a partially identified set (Manski 2003, Tamer 2010), though it is not an identified set in their sense (Section 10).

We then ask two one-world analysts for an interval. A one-world analyst works at one parameter point and one panel. The first analyst ignores interference (Foil 1), while the second models exposure on a randomized arm (Foil 2). We compare each interval with its own image. Because the targets are planted, the true effect is known, and the test can grade the analysts.

Headline.

> Foil 1 is F-OVERCONF at both diagnostic points; no robust Foil 2 verdict is claimed at either point.

The analyst who ignores interference reports a tight interval that sits inside the image and misses the truth. The analyst who models exposure passes the predeclared average-variance gate, but its interval is wide and gets no robust verdict.

The model is a village of 800 people on a fixed random graph, and each person has a preventive-behavior level in $(0, 1)$. A campaign seeds 10% of the people. Section 8 gives the numbers.

1 The planner's problem

A planner compares assignment rules (policies) that act on people who influence each other. Under interference, the outcome of person i depends on the whole assignment vector, so the effect of treating person i alone is not defined. The natural estimand is a contrast of rules,

$$\theta(\pi, \pi'; c) \;=\; \mathbb{E}_c\big[\bar{Y}_T(\pi)\big] \;-\; \mathbb{E}_c\big[\bar{Y}_T(\pi')\big]\,. \tag{1}$$

Here c indexes a complete data-generating process (a kernel), and the expectation is over the simulation noise that remains once c is fixed.

The planner does not know c. She has a class of kernels $\{f(\cdot; c) : c \in C\}$ and a short vector of moments z that she wants to match. History matching then *rules out* parameter points; it does not give a posterior over c. The decision-relevant object is not the kept set in parameter space but the image of θ over that set.

Two sources of doubt sit side by side, and we keep them apart. The first is calibration: which parameter points remain. The second is the analyst: a default estimator fits one panel at one parameter point. For each analyst we report an interval and compare it with its own image.

2 Objects

World, kernel and panel. The world is the graph A plus the covariates X. It is drawn once, and every simulation shares it. The kernel f maps a parameter point c, simulation noise ω and an assignment rule π to a panel (Y_T, D). The kernel is complete, in that it can run every arm on the same noise, and the incompleteness lives in c, not in unobserved counterfactuals inside f (Section 3). The code implements the kernel as vectorized algebra (`simulate_panel` in `kernel.py`).

One run of the kernel gives one panel,

$$\mathcal{D}(c, \omega, \pi) \;=\; \big\{(Y_{iT},\, D_i,\, X_i,\, A_{i\cdot})\big\}_{i=1}^{N}. \tag{2}$$

A parameter point is not a panel: one panel is one draw of ω. The kernel truth $\theta(c)$ uses $R = 200$ paired reps, whereas a one-world analyst sees one panel.

Box. $C \subset \mathbb{R}^5$ is the a priori range of the free parameters $c = (\alpha, \beta, \gamma, \kappa, \sigma_\varepsilon)$. The vector η is fixed inside the kernel and is not in C.

Estimands. Let $\bar{Y}_T$ be the mean of Y_T over people. The code defines three estimands:

$$\theta_1(c) = \mathbb{E}_c[\bar{Y}_T \mid \pi_1] - \mathbb{E}_c[\bar{Y}_T \mid \pi_0], \tag{3}$$

$$\theta_2(c) = \mathbb{E}_c[\bar{Y}_T \mid \pi_2] - \mathbb{E}_c[\bar{Y}_T \mid \pi_0] \quad \text{(headline)}, \tag{4}$$

$$\psi(c) = \theta_2(c) - \theta_1(c) \quad \text{(the premium)}. \tag{5}$$

The arm π_0 seeds no one (control), the arm π_1 seeds 80 people at random (random arm), and the arm π_2 seeds the 80 people with the highest degree (degree arm). The expectation is over simulation noise and over the random assignment in π_1. The code replaces each expectation with a mean over 200 paired reps, and we define θ on the terminal mean $\bar{Y}_T$, not on the path.

Two exposure contrasts, $\delta_{\text{spill}}(c)$ and $\delta_{\text{dir}}(c)$, are defined with the foils (Section 5.7). The letter τ is only the threshold, and the premium is ψ.

Moments and targets. $M : C \to \mathbb{R}^3$ is the map from a parameter point to three simulated summaries. The targets $z \in \mathbb{R}^3$ are the values that $M(c)$ must match. With five free parameters and three moments, the model is underidentified on purpose: the question is which policy effects remain possible, not which c is true.

Implausibility.

$$I(c) \;=\; \big(z - \widehat{M}(c)\big)^{\top} V(c)^{-1} \big(z - \widehat{M}(c)\big), \qquad V(c) = \mathsf{diag}\big(\sigma_z^2 + \sigma_{\mathsf{MC}}^2(c) + \delta^{\circ 2}\big). \tag{6}$$

$I(c)$ is a variance-weighted sum of squared gaps, not a distance. The three terms in V are the Monte Carlo variance of the targets, the Monte Carlo variance of the candidate

moment, and a model-error allowance, which we call the discrepancy δ.

NROY. In the code, NROY is a finite set: the union, over three Latin-hypercube waves, of the sampled points c with $I(c) \leq \tau$. It approximates the level set $\{c \in C : I(c) \leq \tau\}$ but does not enumerate it. Membership means we cannot rule out c. It does not mean that c equals the truth. NROY is not a posterior.

Images.

$$\Theta_{\text{hm}} = \{\theta_2(c) : c \in \mathcal{S}\}, \qquad \Theta_{\text{exp}} = \{\delta_{\text{spill}}(c) : c \in \mathcal{S}\}, \tag{7}$$

where $\mathcal{S}$ is a declared uniform sample of 400 NROY points. The code reports each image as the range $[L, U]$ over $\mathcal{S}$: the declared empirical image. The range is not a distribution. Different NROY points can give different values of θ_2, but they need not.

Diagnostic points. The diagnostic points are the min-I point and the median-I point of NROY, and we write c_0 for either. Both depend on I only, never on θ.

Foils. A foil is a one-world analyst who fits one panel at a diagnostic point c_0. Each foil is judged against its own image and against the truth at c_0 (Section 6). A foil never sees NROY, so it is not a candidate for Θ_{hm}.

3 Potential outcomes, and what counts as data

Under SUTVA, $Y_i = Y_i(D_i)$ with $D_i \in \{0, 1\}$. One of the two numbers per person is missing. That missing number is the incompleteness in the data. A regression with robust standard errors quantifies within-point uncertainty under the analyst's stated assumptions. It is a ruler, not a villain.

This project drops SUTVA: peers influence each other through outcomes ($\gamma > 0$), so $Y_i = Y_i(\mathbf{d})$ with $\mathbf{d} \in \{0, 1\}^N$. The contrast $Y_i(1) - Y_i(0)$ is not defined unless the assignment of everyone else is named. The well-defined object is an overall policy contrast in the sense of Hudgens and Halloran (2008), namely $\theta(\pi, \pi'; c)$ from Section 1.

In the field, the other values of $\bar{Y}(\pi)$ are missing because there is one campaign. In the simulator, f is complete: run both arms on paired noise and subtract. What remains unknown is which $c \in C$ is right, so Θ_{hm} is a not-ruled-out image. It is not a Manski identified set, and it is not a set of possible $Y_i(0)$. It shares the *reporting logic* (do not replace a set by a convenient point) but not the theorem.

Only two objects have empirical analogs.

empirical analog	simulator object	who uses it
micro data of one campaign	one panel $\mathcal{D}(c, \omega, \pi)$	Foil 1, Foil 2
calibration targets	z with its variances and δ	$I(c)$, NROY

The following objects are not data: the Latin-hypercube points, the Monte Carlo reps, the NROY points, the image sample and $\theta(c)$. They are probes of f, and NROY points are arguments of $\theta(\cdot)$, not extra surveys.

4 Order of operations

The order matters: if θ decides which c to keep, the image is a fit, not a report. For that reason the code fixes every exogenous parameter in `config/` before any wave runs.

1. Fix the kernel f, the box C, the targets z, the discrepancy δ, the threshold τ, the estimands and the rule for the diagnostic points.
2. Draw the world (A, X) once.
3. History matching. Evaluate $I(c)$ on a space-filling design and keep the points with $I \leq \tau$. Later waves search the box of the kept points; NROY is the union of the kept points. An empty NROY is a finding: the pair (f, z) is rejected. Never move τ or δ.
4. Leg A. Sample 400 NROY points, compute $\theta_2(c)$ and $\delta_{\text{spill}}(c)$ at each, and report Θ_{hm} and Θ_{exp} as ranges. Then evaluate the endpoint candidates again with fresh reps.
5. Leg B. At each diagnostic point, fit the two foils on one panel and compare each interval with its own image and with the kernel truth at that point.

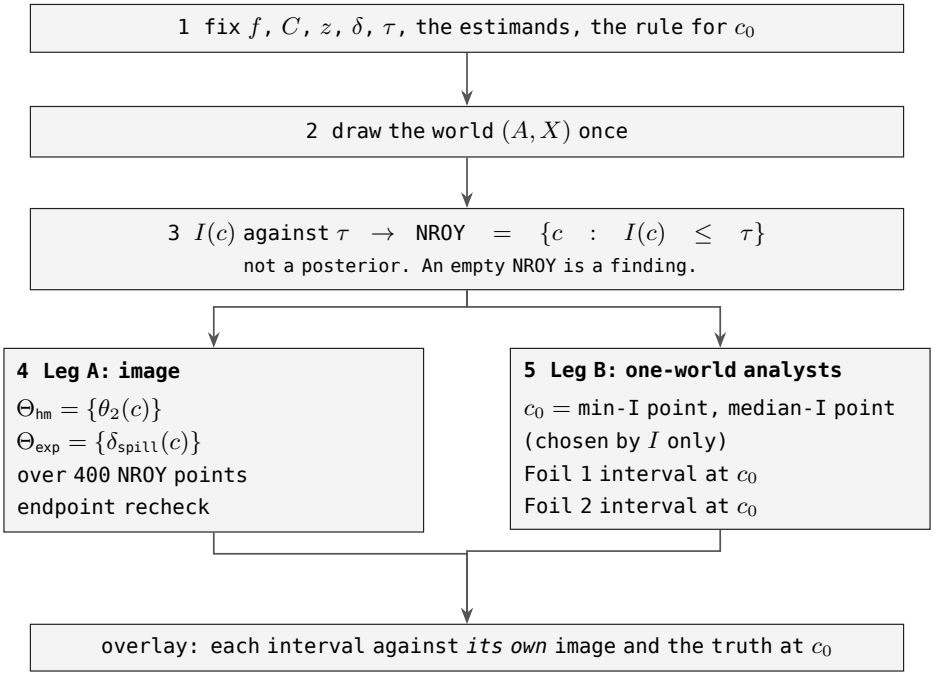


Figure 1: Order of operations. Legs A and B are two reports and do not merge into one object.

5 Design of exogenous quantities

The code fixes the history-matching quantities below before the waves run, whereas the later Foil 1 rescaling is post hoc and is disclosed in the results.

5.1 Threshold τ. The code uses the 0.95 quantile of χ^2 with three degrees of freedom, $\tau = 7.8147$, as its design threshold. This choice does not assert that the implemented $I(c)$ has an exact χ^2 sampling law or define a credible region. It is a rule for ruling out. If we move τ after we see NROY, a filter becomes a fit. The same level, 95%, sets both foil intervals.

5.2 Discrepancy δ. The code adds the fixed discrepancy $\delta = (0.02, 0.05, 0.02)$ to V. In this planted exercise it is a design tolerance, not an estimated model-error variance. With external targets, the same slot is the allowance for a model that is not the world. If we loosen δ after a wave, it is the same error as moving τ.

5.3 The box C. The box is wide enough for underidentification ($5 > 3$) but also narrow: persistence, peer weight and the effect of seeding are positive, and the intercept does not push Y to saturation for every c. The value $\gamma = 0$ is outside the box on purpose, since the project is about interference. A world without interference that survives calibration is a separate finding.

5.4 How c is drawn. No one picks a favorite parameter point by looking at θ. Wave 1 is a Latin hypercube of 1000 points in C, and the code keeps each point with $I(c) \leq \tau$. Waves 2 and 3 draw 1000 new points in the bounding box of the points kept in the previous wave. The box widens by 20% of its span and is clipped to C. There is no emulator: the code simulates every point. NROY is the *union* of the kept points over all waves. The planted point $c^\star$ is the center of C, and we chose it without looking at θ.

5.5 Targets z. The targets are planted: $z := \widehat{M}(c^\star)$, with 200 reps and its Monte Carlo variances. The file `data/planted.json` stores them. Planting supplies a known benchmark and makes a match plausible. It does not guarantee that a finite Latin-hypercube search keeps a point. It also makes the true effect known, and that is the only way to check an overlay.

There are three moments. They have legacy names from an earlier version that used World Bank data. The code keeps the names so that saved files keep their keys. They do not describe immunization or internet use here.

slot (legacy name)	definition	arm
level_imm	$\mathbb{E}[\bar{Y}_T \mid \pi_2]$	degree arm
gradient_edu	$\mathrm{corr}(Y_T, x_1 \mid \pi_2)$	degree arm
level_internet	$\mathbb{E}[\bar{Y}_T \mid \pi_1]$	random arm

The moments use $R_m = 30$ paired reps and run two arms, not three. The control arm is not simulated for the moments. So θ_2, a contrast against control, is not a moment that the

code matches. The code uses fewer reps for the moments than for the estimands because the search must be cheap. The estimands are computed only for kept points. The targets and estimands each use 200 reps. Equal replication counts do not imply equal Monte Carlo precision across these different statistics.

5.6 Diagnostic points. The min-I point is the closest match in NROY. The median-I point is the point at position $\lfloor |\text{NROY}|/2 \rfloor$ in the order of I, and both depend on moment fit, not on the estimand. We use two points, not 400. Running an analyst at every kept point gives the analyst the calibration set, and default practice does not do that.

5.7 Foils.

Foil 1: the SUTVA analyst. Foil 1 uses the degree arm and one panel. It fits OLS of Y_T on $Z_i = (1, D_i, X_i^\top)$, which has $k = 5$ columns, and the coefficient on D is $\hat{\beta}$. The standard error is HC1:

$$\widehat{\text{Var}}_{\text{HC1}}(\hat{\beta}) \;=\; \frac{n}{n-5}\,(Z^\top Z)^{-1}\Big(\sum_{i=1}^{n} \hat{u}_i^2\, Z_i Z_i^\top\Big)\,(Z^\top Z)^{-1}. \tag{8}$$

The interval for $\hat{\beta}$ is $\hat{\beta} \pm 1.96\,\text{SE}$. On the degree arm, D is "being a top-degree person." The SUTVA analyst turns $\hat{\beta}$ into a policy estimate for the whole village, $(k/N)\hat{\beta}$, with $k/N = 0.1$, and the interval scales the same way. We call this interval J_1, and we compare J_1 with Θ_{hm}. The scaling by a known positive constant keeps the analyst's own coverage statement, but it does not repair the analyst's assumptions.

Foil 2: Hájek on the random arm. Foil 2 uses the random arm and one panel. The random arm has a known lottery, so we can write exposure probabilities before we see Y. The exposure share is

$$k_i = \max\Big(\sum_j A_{ij},\, 1\Big), \qquad s_i = \frac{\sum_j A_{ij} D_j}{k_i}. \tag{9}$$

The share s_i counts treated neighbors, not neighbor outcomes, and an isolated person has $k_i = 1$ and $s_i = 0$. The code defines four strata:

$$\begin{aligned} S_1 &= \{i : D_i = 1\}, \\ S_{0a} &= \{i : D_i = 0,\ s_i = 0\}, \\ S_{0b} &= \{i : D_i = 0,\ 0 < s_i < 0.25\}, \\ S_{0c} &= \{i : D_i = 0,\ s_i \geq 0.25\}. \end{aligned}$$

In a panel, if any stratum has fewer than 20 people (a thin cell), the code merges S_{0b} into S_{0c} for that panel. The kernel truth uses the realized stratum means of the true Y, averaged over 200 random-arm reps:

$$\delta_{\text{spill}}(c) = \overline{Y}_{S_{0c}} - \overline{Y}_{S_{0a}}, \qquad \delta_{\text{dir}}(c) = \overline{Y}_{S_1} - \overline{Y}_{S_{0a}}. \tag{10}$$

Foil 2 replaces each stratum mean with a Hájek mean, which weights each person by the inverse of the exposure probability,

$$\widehat{\mu}_S \;=\; \frac{\sum_i \mathbf{1}\{i \in S\}\, Y_i / p_{i,S}}{\sum_i \mathbf{1}\{i \in S\} / p_{i,S}}. \tag{11}$$

The first-order probabilities $p_{i,S}$ are exact: under the random arm, the number of treated neighbors of an untreated person of degree d is hypergeometric. The variance is the estimator of Aronow and Samii (2017). Its joint exposure probabilities come from 10,000 simulated random-arm assignments, and Young's bound covers pairs that never appear together. The interval is the estimate ± 1.96 SE. We call it J_2, and we compare it with Θ_{exp}.

The exposure map uses neighbor treatment, whereas the kernel peer term uses neighbor outcomes and the effect travels more than one step. So the map is not exact for the kernel. This limits Foil 2, but it is not a reason to replace Foil 2 with a second SUTVA analyst. Foil 2 does not estimate θ_2: the degree arm has no lottery, so given A its assignment probability is 0 or 1, and inverse-probability weights are not defined.

6 Interval geometry

Intervals and images are subsets of $\mathbb{R}$. A verdict is an inclusion, not a slogan. Let $J = [lo, hi]$ be the interval of a foil at one diagnostic point, let $\Theta = [L, U]$ be the image of the *same* kind of functional over the 400 sampled NROY points, and write $\text{int}(\Theta) = (L, U)$. The function `classify_interval` gives:

$$\text{F-HONEST} \iff lo \le L \text{ and } hi \ge U \iff \Theta \subseteq J, \tag{12}$$

$$\text{F-OVERCONF} \iff lo > L \text{ and } hi < U \iff J \subset \text{int}(\Theta). \tag{13}$$

Every other case has no verdict, including an interval that overlaps the image with neither inclusion and an interval with an end that is not finite. Each foil goes against its own image: J_1 against Θ_{hm}, and J_2 against Θ_{exp}.

A verdict is a statement about interval geometry. F-OVERCONF means the analyst's interval is narrower than the set of effects that the targets leave open, and it does not say that the interval fails to cover the truth in repeated sampling. The code reports a separate indicator for that, `covers_truth`, and the repeated-assignment check (Section 8.5) tests Foil 2 apart from the verdicts.

7 Illustration: a planted village

One village has $N = 800$ people. The graph has an edge rule rather than a mean degree: the code draws $A_{ij} \sim \text{Bernoulli}(p)$ with $p = 4/N$ for every ordered pair, takes $A \mid A^\top$ and sets the diagonal to zero. An edge therefore appears with probability $1-(1-p)^2 \approx 8/N$, and the mean degree is about 8. The theory gives 7.97, and the graph in this project measures 8.085. The graph is drawn once with a fixed seed and is not calibrated.

$Y_{i,t} \in (0,1)$ is the level of preventive behavior. The covariates are $X \sim U[0,1]^3$: education or literacy x_1, resources x_2 and baseline propensity x_3. A rule seeds $k = 80$ people from $t = 0$, and the three rules are control (π_0), random (π_1) and degree (π_2). Ties in degree go to the lower index. In the degree arm, 35 of the 36 people at degree 12 are seeded, and the arm is deterministic given the graph.

The kernel is a logit linear-in-means update: a linear-in-means index passed through a logistic link. For $t = 0, \ldots, T-1$ with $T = 20$ and $t^* = 0$,

$$Y_i(t+1) \;=\; \Lambda\Big(\alpha + \beta Y_i(t) + \gamma \frac{(A\,Y(t))_i}{k_i} + \eta \cdot X_i + \kappa D_i \mathbf{1}\{t \geq t^*\} + \sigma_\varepsilon\, \varepsilon_i(t)\Big), \tag{14}$$

with $\Lambda(z) = 1/(1+e^{-z})$, $\varepsilon_i(t) \sim N(0,1)$ and $Y_i(0) \sim U[0,1]$. Peers enter through the lagged outcomes of neighbors, not through neighbor treatment. The update rule is chosen, not derived, and its lineage has three parts: Manski (1993) gives the linear-in-means form, Bramoullé, Djebbari and Fortin (2009) give the row-normalized peer mean, and Brock and Durlauf (2001) give the logistic link.

	box / value	role
α	$[-2.5, -0.5]$	intercept of the index
β	$[0.05, 0.60]$	own lag
γ	$[0.05, 0.80]$	weight on the peer mean
κ	$[0.25, 2.50]$	effect of being seeded (the same on every arm)
σ_ε	$[0.01, 0.40]$	standard deviation of the shock
η	$(0.05, 0.10, 0.02)$	covariate weights, fixed, not calibrated

The planted point is the center of C: $c^\star = (-1.5,\, 0.325,\, 0.425,\, 1.375,\, 0.205)$. The targets are $z = \widehat{M}(c^\star) = (0.2638,\, 0.05,\, 0.2618)$ for level_imm, gradient_edu and level_internet. The discrepancy is $\delta = (0.02, 0.05, 0.02)$, and the threshold is $\tau = 7.8147$.

Every random stream is seeded. Rep r of every arm shares one state stream (the start $Y(0)$ and all shocks) and has its own assignment stream, so the reps are paired and contrasts between arms have low noise. The whole pipeline is deterministic, and a rerun reproduces every number.

8 Results

Headline.

> Foil 1 is F-OVERCONF at both diagnostic points; no robust Foil 2 verdict is claimed at either point.

All numbers in this section come from the committed results in `results/`.

8.1 History matching kept a large NROY set. Against the targets $z = \widehat{M}(c^\star)$, three waves kept 202, 416 and 373 points of 1000 each, and the union is $|\text{NROY}| = 991$ (Figure 2). The minimum I in the three waves is 0.0017, 0.0050 and 0.0071, and the median I is 50.37, 11.89 and 13.59. The threshold is $\tau = 7.8147$. The set did not collapse to a point.

The closest NROY point has $I = 0.0017$, but it does not recover $c^\star$: its parameters are $\alpha = -1.4871$, $\beta = 0.399$, $\gamma = 0.3301$, $\kappa = 1.3446$ and $\sigma_\varepsilon = 0.1984$. Fit does not pick the truth. This point is the min-I point by the declared rule, not because it won on θ. The median-I point, by contrast, has $I = 2.0208$ and a much larger κ (2.4654).

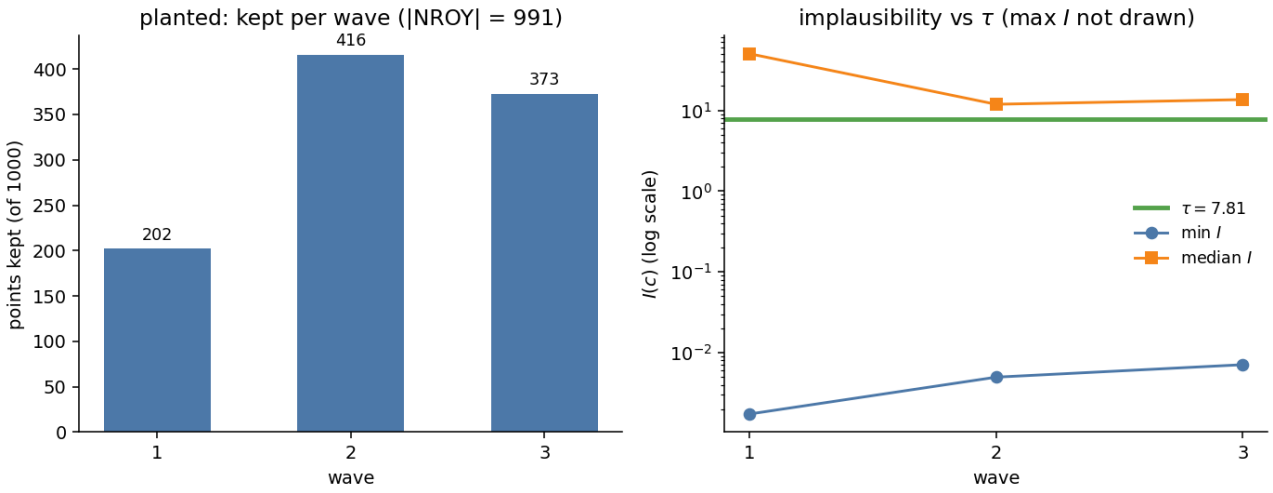


Figure 2: Planted history matching. Left: points kept in each wave. Right: min and median $I(c)$ against the threshold τ (log scale). Later waves concentrate on the kept region. The threshold does not move.

8.2 The image of θ_2 is not a point. At the planted point, $\theta_1(c^\star) = 0.03631$ and $\theta_2(c^\star) = 0.03826$, and the premium is $\psi(c^\star) = 0.00195$. In this model, seeding the highest-degree people barely beats seeding at random, but that is a fact about this $c^\star$, not a theorem about targeting. The spillover at $c^\star$ is $\delta_{\text{spill}}(c^\star) = 0.00792$.

Over the 400 sampled NROY points, the headline image is $\Theta_{\text{hm}} = [0.00587, 0.07325]$, and the image of the spillover is $\Theta_{\text{exp}} = [-0.000571, 0.026893]$. The honest report for θ_2 is the range, not 0.03826. The range is not a sampling distribution and not a posterior.

Figure 3 shows every sampled point. Each bar is a pointwise Monte Carlo SE at a fixed c. For θ_2 the bars are smaller than the dots. For δ_{spill} they are visible.

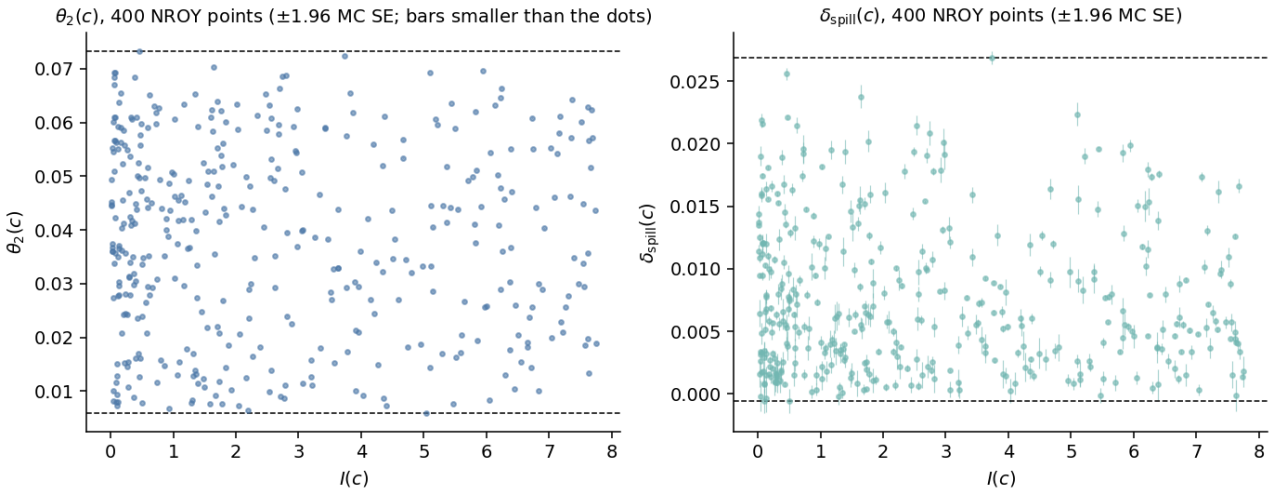


Figure 3: The 400 sampled NROY points against $I(c)$. Left: $\theta_2(c)$. Right: $\delta_{\text{spill}}(c)$. Bars show ± 1.96 pointwise MC SE. Dashed lines mark the image endpoints.

The endpoint recheck. The code evaluates again each point that sets an endpoint. It uses 200 fresh reps on separate streams. Table 1 gives the result.

Table 1: Independent recheck of the endpoint candidates (SE is the pointwise MC SE).

endpoint	NROY index	pilot (SE)	fresh (SE)
Θ_{hm} lower	877	$+0.005870$ (0.000004)	$+0.005871$ (0.000003)
Θ_{hm} upper	802	$+0.073250$ (0.000006)	$+0.073255$ (0.000006)
Θ_{exp} lower	944	-0.000571 (0.000502)	$+0.000201$ (0.000491)
Θ_{exp} upper	904	$+0.026893$ (0.000259)	$+0.027274$ (0.000263)

Each bar in `image_points.json` is a pointwise Monte Carlo SE conditional on a fixed parameter point c. The displayed image endpoints are extrema selected from 400 noisy estimates. So the SE attached to the extremizing point is not an SE for the endpoint. It does not include extremum-selection uncertainty. Fresh independent re-evaluation of the pilot-selected extremizers removes reuse-of-noise bias for those candidates. It is not simultaneous inference for the true extrema.

8.3 The overlay. Figure 4 puts each interval next to its own image at the two diagnostic points. The rows are the min-I point and the median-I point. The left column shows θ_2 against Θ_{hm} with the Foil 1 interval, and the right column shows δ_{spill} against Θ_{exp} with the Foil 2 interval. A star marks the truth at that point, and a triangle marks the truth at $c^\star$.

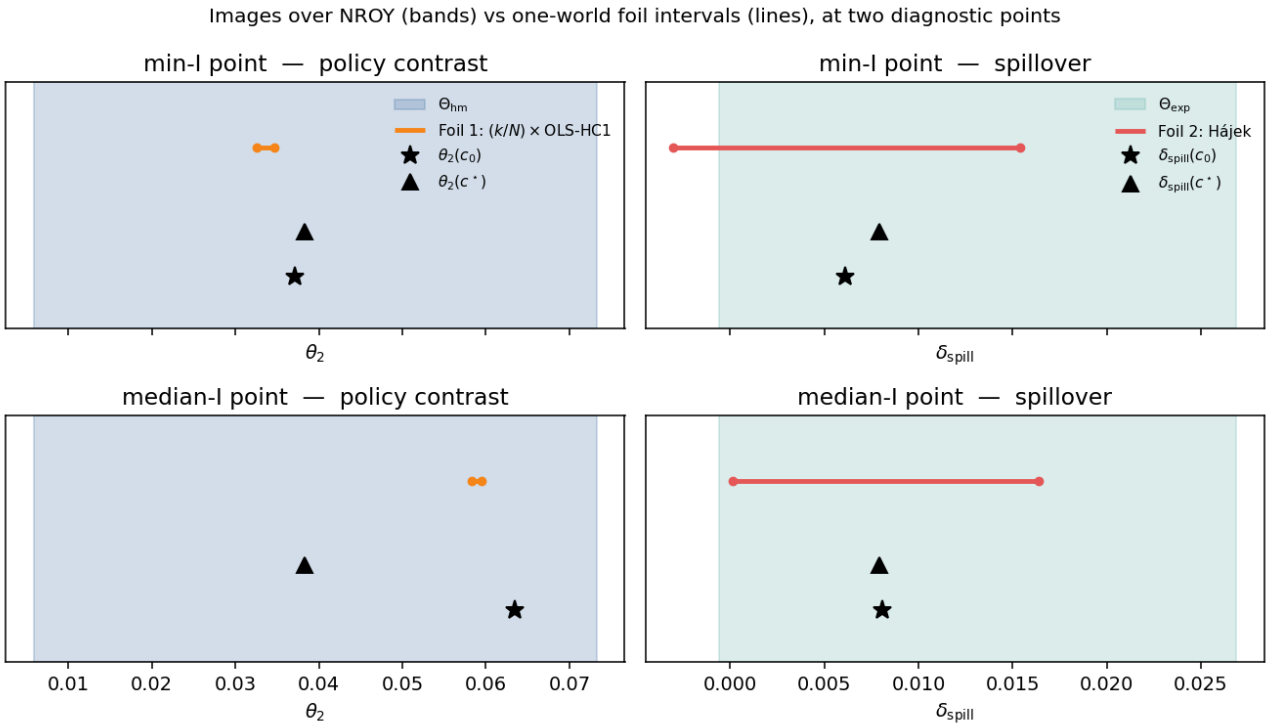


Figure 4: Images and one-world intervals at the two diagnostic points. Left: θ_2 against Θ_{hm} with Foil 1. Right: δ_{spill} against Θ_{exp} with Foil 2. Each interval is compared with its own image.

Table 2: The two foils at the two diagnostic points.

	min-I point ($I = 0.0017$)	median-I point ($I = 2.0208$)
truth $\theta_2(c_0)$	0.03708	0.06346
Foil 1 interval J_1	$[0.03259, 0.03465]$	$[0.05832, 0.05949]$
Foil 1 verdict	F-OVERCONF	F-OVERCONF
Foil 1 contains the truth	no	no
truth $\delta_{\text{spill}}(c_0)$	0.00613	0.00809
Foil 2 estimate (SE)	0.00619 (0.00470)	0.00826 (0.00416)
S_{0b} merged	no	no
Foil 2 interval J_2	$[-0.003022, 0.015410]$	$[0.000112, 0.016406]$
Foil 2 verdict (pilot image)	no verdict	F-OVERCONF
Foil 2 contains the truth	yes	yes

8.4 Foil 1: the SUTVA analyst. Table 2 shows that the Foil 1 interval is narrow. At the min-I point, $\hat{\beta} = 0.3362$ with interval $[0.3259, 0.3465]$, and at the median-I point, $\hat{\beta} = 0.5890$ with interval $[0.5832, 0.5949]$. The scale factor is $k/N = 0.1$, and five statements stay with every report of Foil 1.

1. Foil 1 is not an identified estimator of the simulator's θ_2 under the actual interference model. It is the whole-village policy estimate implied by a SUTVA analyst, $(k/N)\hat{\beta}$, with $k/N = 0.1$.

2. That interpretation assumes consistency/SUTVA, conditional exchangeability and positivity given X, a correctly specified additive constant-effect outcome model, and a sampling/asymptotic basis for HC1. The deterministic top-degree assignment does not establish those assumptions.

3. After scaling, the Foil 1 intervals are strictly inside Θ_{hm} and exclude $\theta_2(c_0)$ at both diagnostic points. So F-OVERCONF is an assumption-contingent interval-geometry finding. It is not a coverage claim for θ_2 under interference.

4. Across 200 paired panels, about 90% of the gap between θ_2 and (k/N) times the 200-panel mean OLS coefficient is the policy effect on unseeded people. For the displayed replication-0 point estimate, the corresponding shares are about 96% at min-I and 92% at median-I.

5. The rescaling was adopted after the frozen numbers were inspected. Its justification is the SUTVA aggregation identity. The post hoc design correction is disclosed.

The gap decomposition. With paired streams, the policy effect splits exactly into a seeded part and an unseeded part. Let S be the seeded set, and let $y_2 - y_0$ be the effect on a person in one rep. Then

$$\theta_2 = \underbrace{\tfrac{1}{N}\textstyle\sum_{i\in S}(y_2 - y_0)}_{\text{the seeded part}} + \underbrace{\tfrac{1}{N}\textstyle\sum_{i\notin S}(y_2 - y_0)}_{\text{the unseeded part}}. \tag{15}$$

The unseeded part is zero under SUTVA. The seeded policy contrast is the mean of $y_2 - y_0$ over S, and under interference it is not a SUTVA effect of own treatment. Table 3 gives the numbers. The gap for the mean beta averages over 200 panels, whereas the gap for the displayed beta uses the one panel that Foil 1 sees. We keep the two apart.

Table 3: Gap decomposition for Foil 1 (200 paired reps).

	min-I point	median-I point
θ_2	0.03708	0.06346
seeded part	0.03374	0.05925
unseeded part	0.00333	0.00421
seeded policy contrast	0.3374	0.5925
mean beta	0.3339	0.5880
gap for the mean beta	0.00369	0.00466
unseeded share of that gap	0.9034	0.9027
OLS-error part of that gap	0.00036	0.00045
gap for the displayed beta	0.00346	0.00456
unseeded share of that gap	0.9642	0.9236
baseline gap (seeded minus unseeded, Y under control)	0.0007	0.0007

Most of the gap is the effect on people who were not seeded, and a SUTVA analyst has no term for it.

Why there is no forest foil. This was decision D3. In the degree arm, treatment is a deterministic function of degree and the index tie-break. Conditioning on degree leaves severe practical positivity failure: no treated people appear below degree 12, no untreated people appear above degree 12, and only one untreated person appears at degree 12. Omitting degree leaves the assignment variable unadjusted. A forest can differ numerically from OLS, but it retains the same SUTVA blind spot and does not identify θ_2 from this panel. A design-based third foil needs randomization across villages (Hudgens and Halloran 2008, Baird et al. 2018).

8.5 Foil 2: Hájek on the random arm. At each diagnostic point, covers_truth reports whether the displayed interval contains the simulator truth. It is a descriptive containment indicator for that realized panel, not an estimate of repeated-sampling coverage and not a validity gate. Foil 2 is checked separately by the pre-declared repeated-assignment calibration in scripts/check_foil2_coverage.py. Both Foil 2 intervals contain the truth (Table 2).

Against the saved 200-rep pilot image, $\Theta_{\text{exp}} = [-0.000571, 0.026893]$, the median-I Foil 2 interval $[0.000112, 0.016406]$ is strictly interior and is therefore classified F-OVERCONF. That classification is not robust to endpoint Monte Carlo noise. An independent 200-rep recheck of the pilot-selected lower-end candidate is 0.000201 (pointwise MC SE 0.000491). This value is above the lower end of the interval. The recheck evaluates selected candidates. It neither identifies nor bounds the true extrema over all 400 sampled NROY points. We therefore claim no robust Foil 2 verdict at either diagnostic point. At the min-I point, the interval reaches below the lower end of Θ_{exp}, so there is no verdict at all.

An earlier version (commit ebed7ce) used a first-order variance that dropped the covariance between the strata, and commit 36ed3e6 replaced it with the Aronow–Samii variance.

The calibration check. The script scripts/check_foil2_coverage.py is a required check. Its gate uses outcomes that obey the exposure map exactly, with a true spillover of 0.03, and it draws 10,000 random-arm assignments. The gate rule was written before the run: the gate fails if the ratio of the mean estimated variance to the empirical variance, plus 1.645 bootstrap SE, is below 1. The gate passes, with a mean estimate of 0.03006 and a ratio of 1.034 (bootstrap SE 0.0154). The result:

> In a fixed-world repeated-assignment check whose outcomes obey the exposure map exactly, the nominal 95% Foil 2 Wald interval covered 93.4% of assignments (exact 95% binomial interval 92.9%–93.9%). The average estimated variance was 1.034 times the empirical design variance, so the shortfall is not explained by average variance underestimation. It reflects the finite-sample behavior of this studentized normal interval, with the implemented Hájek linearization and simulated joint exposure probabilities.

The check is specific to this implementation. It is not a test of the Aronow–Samii theorem.

Their exact result is for Horvitz–Thompson totals with exact joint probabilities. The code uses Hájek means, a first-order linearization and one simulated table of joint probabilities.

The script also reports the kernel, where the exposure map is not exact because spillover also travels two steps. Over 200 kernel panels, the coverage of the nominal 95% interval is 0.920 at $c^\star$, 0.930 at the min-I point and 0.925 at the median-I point. If we redraw only the assignment and hold the noise fixed, the coverage is 0.930, 0.930 and 0.9267. The code reports these numbers but does not gate on them.

8.6 The World Bank run: F-EMPTY. The command `run_poc.py worldbank` reproduces the August 2026 attempt with real World Bank anchors. It is a historical sensitivity run, and the anchors are three cross-country indicators. The targets are $z = (0.8621, 0.6054, 0.6864)$. The run ends in F-EMPTY: wave 1 keeps no point, and none of the 1,000 sampled wave-1 points passed the threshold. The best I is 204.35, the median is 1093.22 and the maximum is 1818.34, against $\tau = 7.8147$, and the code does not loosen τ or δ. The closest sampled moments are $(0.6188, 0.0545, 0.6133)$, and the closest sampled education correlation was 0.05, against a target of 0.61 (Figure 5). The finding is about the pair (kernel, these anchors). It is not a wide image.

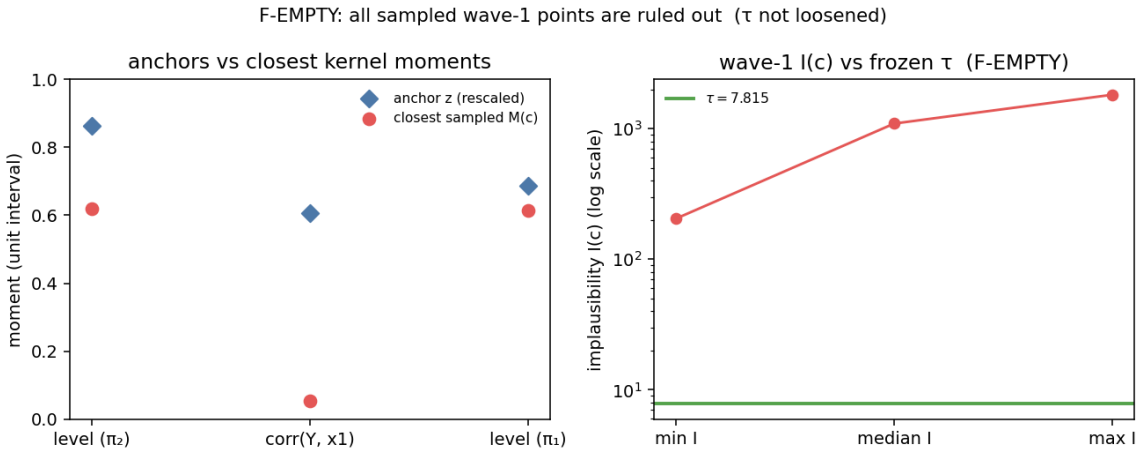


Figure 5: F-EMPTY on the World Bank anchors. Left: anchors against the closest sampled moments. Right: wave-1 $I(c)$ against τ (log scale).

The World Bank sensitivity run reproduces a historical F-EMPTY calculation. Its diagonal weights are legacy heuristic cross-entity dispersion terms, not design-based sampling variances or village-level measurement-error variances. The source includes overlapping regional/income aggregates, uses ddof=0 for level dispersion, and uses an asymptotic n-2 correlation convention.

9 Who will care, and who will not

Novelty at the level of slogans is weak. History matching already gives NROY in input space (Craig, Goldstein and Vernon). Equifinality of agent-based models is not new (Beven). Partial identification already says: report the set. Interference already has overall and direct effects (Halloran, Hudgens and Halloran). A regression with robust standard errors already gives an interval for a given panel. None of these pieces is new, so the remaining wedge is the composition: a named policy contrast under interference is reported as the image of a history-matched set, and two one-world analysts are set beside it. The simulator is planted, so the truth is known. That is enough for an applied project, but it is not a methods paper.

Anyone who calibrates a simulator for policy. This matters if you publish a policy number from an agent-based model, a structural model or a microsimulation. Today you probably report the Monte Carlo error at one best parameter point. The overlay shows a split. Sampling error at one point is not uncertainty about which points the targets still allow. An empty sampled NROY is a usable negative result for the declared design: this run found no sampled point that passed the threshold for those targets. It does not establish infeasibility over all of C. You must not loosen the threshold. The width of the image is measured, not assumed. Here $\Theta_{\mathsf{hm}} = [0.00587, 0.07325]$.

Empirical causal inference. This is not a new identification strategy: it does not replace unconfoundedness, an instrument, difference-in-differences or a regression discontinuity, and it does not estimate effects that vary by covariate on a network. It does, however, matter in two narrow cases.

(1) You use a model as the identification vehicle, and you quote one $\hat{\theta}$. The model can be structural, an epidemic simulator or an agent-based model of a campaign. The object to report is then the declared empirical image of θ over the sampled NROY points. A one-world interval quantifies within-point uncertainty under the analyst's stated assumptions. It is not the image.

(2) Interference is in the model, and the individual effect is not defined. The well-posed object is a contrast of assignment rules. A SUTVA analyst on a targeting arm, where D is a function of the graph, measures something near "being a top-degree person." That is an identification remark, not an estimator contest.

This project, specifically. The overlay is checkable because the targets were planted. The one attempt at external targets (World Bank) ended in F-EMPTY, so this is not an empirical estimate of any real campaign, and we do not present it as one. It shows that the composition can be run, with declared verdicts, including F-EMPTY and no verdict. As a contribution to empirical causal inference on real networks, it is a toy with a clean dictionary of objects. It becomes more only if someone supplies targets that this kernel can match, and those targets need a stated target population and a transport model. The threshold τ must not move.

10 What the procedure is not

- Not a posterior over c, and not a distribution of treatment effects. Θ_{hm} comes from a thresholded discrepancy, in which sampling noise is folded into V and the kernel is taken as true.
- Not a Manski or Tamer identified set. This is a finite empirical image produced by thresholding simulated moment discrepancies for a fully specified kernel. The analogy is only the reporting rule: do not collapse a set-valued result to one convenient point.
- Not a guarantee that the image is wide: its width is measured, not assumed.
- Not a claim that the closest NROY point recovers $c^\star$, which it does not: fit does not pick the truth.
- Not a coverage claim for Foil 1: F-OVERCONF is about interval geometry under the analyst's assumptions.
- Not a finding that one-world analysts are bad at networks. Each quantifies within-point uncertainty under its own stated assumptions. The overlay asks whether one parameter point is enough.
- Not a claim about the real world: the project has one planted point, one graph and one set of targets, and it claims no external validity.

Implementation. Repo `ecn593-applied-project`, branch `main`. `scripts/replicate.py` reruns everything. Each result file records the commit that produced it (`git_commit`). $|\text{NROY}| = 991$, $\Theta_{\text{hm}} = [0.00587, 0.07325]$, $\theta_2(c^\star) = 0.03826$. Foil 1 is OLS-HC1 scaled by k/N. Figures are redrawn from the saved results. `README.md` lists the commands to replicate every number.